Module 1 - Random Variables and Processes:

Introduction, Probability, Conditional Probability, Random variables. Statistical Averages: Function of a random variable, Moments, Random Processes, Mean, Correlation and Covariance function: Properties of autocorrelation function, Cross–correlation functions, Gaussian Process: Gaussian Distribution Function.

[Text 2: 5.1, 5.2,5.3,5.4,5.5,5.6,5.9] RBT: L1, L2

Module 2 - Amplitude Modulation Fundamentals:

AM Concepts, Modulation index and Percentage of Modulation, Sidebands and the frequency domain, AM Power, Single Sideband Modulation.

[Text1: 3.1, 3.2,3.3,3.4,3.5] RBT: L1, L2, L3

AM Circuits: Amplitude Modulators: Diode Modulator, Transistor Modulator, collector Modulator. Amplitude Demodulators: Diode Detector, Balanced Modulators: Lattice Modulators.

[Text1: 4.2,4.3,4.4,] RBT: L1, L2, L3

Frequency Division Multiplexing: Transmitter-Multiplexer, Receiver-Demultiplexer.

[Text1: 10.2] RBT: L1, L2, L3

MODULE-3 - Fundamentals of Frequency Modulation:

Basic Principles of Frequency Modulation, Principles of Phase Modulation, Modulation index and sidebands, Noise Suppression Effects of FM, Frequency Modulation versus Amplitude Modulation.

[Text1: 5.1,5.2,5.3,5.4,5.5] RBT: L1, L2, L3

FM Circuits: Frequency Modulators: Voltage Controlled Oscillators. , Frequency Demodulators: Slope Detectors, Phase Locked Loops.

[Text1: 6.1,6.3] RBT: L1, L2, L3

Communication Receiver: Super heterodyne receiver, Frequency Conversion: Mixing Principles, JFET Mixer. [Text1: 9.2,9.3] RBT: L1, L2, L3

Module 4 - Digital Representation of Analog Signals:

Introduction, Why Digitize Analog Sources?, The Sampling process, Pulse Amplitude Modulation, Time-Division Multiplexing, Pulse Position Modulation: Generation and Detection of PPM wave. The Quantization Process. Pulse Code Modulation: Sampling, Quantization, Encoding, line Codes, Differential encoding, Regeneration, Decoding, filtering, multiplexing. [Text2: 7.1,7.2,7.3,7.4,7.5,7.6,7.8,7.9] RBT: L1,L2,L3

Module 5 Baseband Transmission of Digital signals:

Introduction, Inter symbol Interference (ISI), Eye Pattern, Nyquist criterion for distortionless Transmission, Baseband M-ary PAM Transmission.

[Text2:8.1,8.4,8.5,8.6,8.7]

Noise: Signal to Noise Ratio, External Noise, Internal Noise, Semiconductor Noise, Expressing Noise Levels, Noise in Cascade Stages.

[Text1:9.5] RBT:L1,L2,L3